

Survey of high-level simulation tools (Oct'11)

Software	Vendor	Typical Applications of the software	Primary Markets for which the software is applied
aGPSS	Beliber AB	General purpose discrete events simulation of situations with uncertainty, requiring many runs	Education, esp. students of business, logistics, supply chain systems
Analytica 4.4	Lumina Decision Systems	Investment,risk,decision,portfolio network flow analysis,systems dynamics,resource,R&D planning,organization simulation	Business, financial, public policy, energy, environmental, healthcare, defense, manufacturing, education, telecommunication
AnyLogic Simulation Software and Services	AnyLogic North America	AnyLogic is the only tool combining System Dynamics, Discrete Event, and Agent Based methods and allowing hybrid combinations	Supply Chain, Healthcare, Process Redesign, Education, Warehouse Manufacturing, Defense
Arena Simulation Software	Rockwell Automation	purpose DES software tool. Any process you can describe can be modeled in Arena.	Manufacturing, 6-Sigma, Packaging, Supply Chain, Airports, Healthcare, Military/Defense, Service, Call Centers, Mining, Ports
Bluesss Simulation System	Stanislaw Raczynski	General purpose simulation system, C++ related	Education, engineering
Capacity Planning Simulator	ProModel Corporation	Manufacturing, Warehouse, Distribution Center, Logisitcs, Supply Chain Capacity Planning	Manufacturing, Logistics, Supply Chain
CSIM 20	Mesquite Software	creating simulation models of systems	computer and communications systems, education, systems research
Enrmgmsuite	Techno Software International	Increase Energy efficiency of system , subsystem, components.	All energy consuming industries (Process / Power / Pharmaceutical /Paper / Steel ...)
Enterprise Dynamics	INCONTROL Simulation Solutions	Enterprise Resource Simulation Platform for process improvement, capacity investigations and investment evaluations	Manufacturing, Airlines & Airports, Railway, Road Transport, Logistics, Material Handling, Crowd Management

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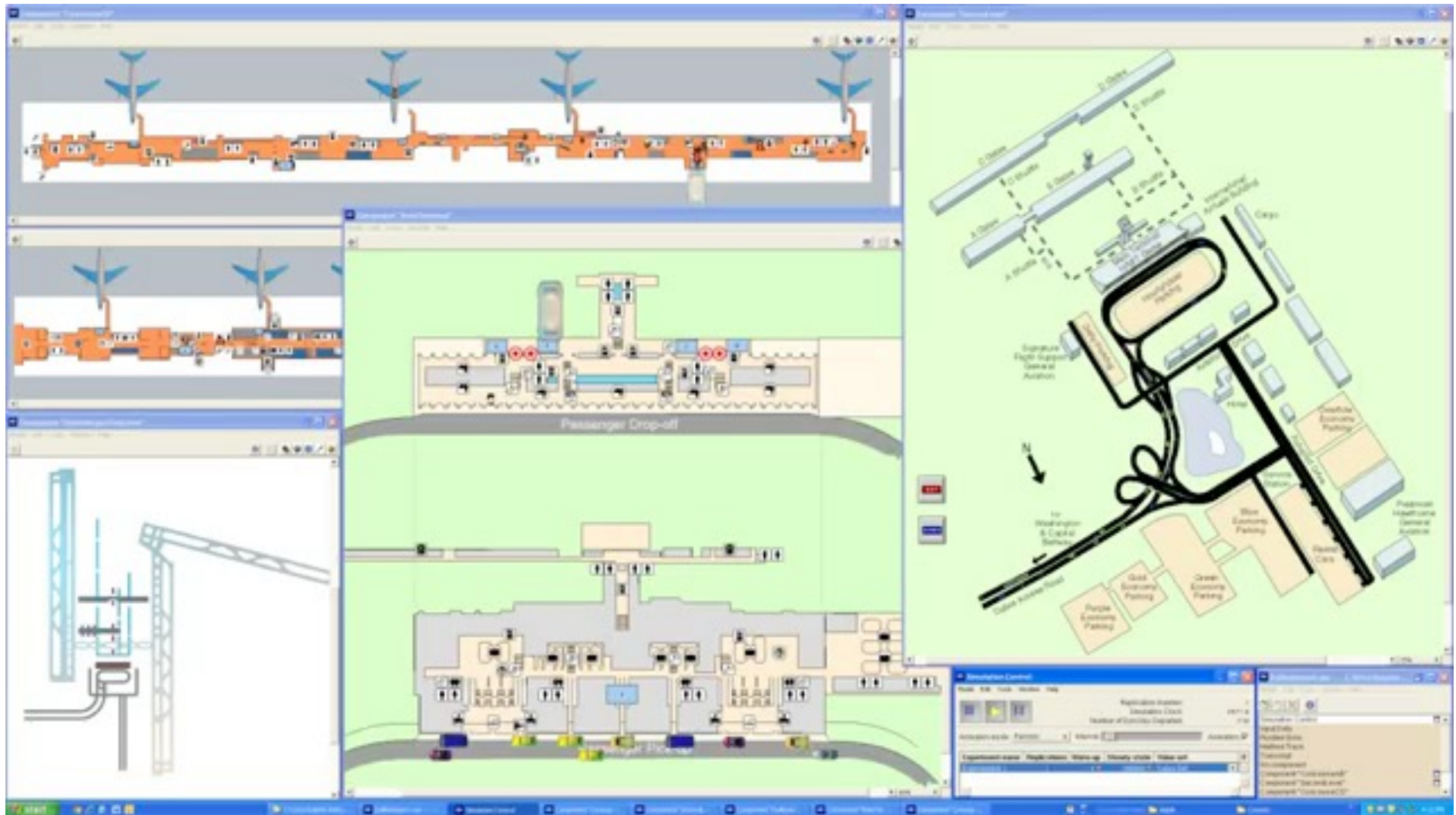
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SIMUL8 Professional	SIMUL8 Corporation	Optimize throughput, maximize resource utilization, identify bottlenecks, reduced risk decisions, business process management	Business processes: call center, manufacturing, supply chain, logistics, healthcare, financial, education
SIMUL8 Standard	SIMUL8 Corporation	Optimize throughput, maximize resource utilization, identify bottlenecks, reduced risk decisions, business process management	Business processes: call center, manufacturing, supply chain, logistics, healthcare, financial, education
SIMUL8 Web	SIMUL8 Corporation	Optimize throughput, maximize resource utilization, identify bottlenecks, reduced risk decisions, business process management	Business processes: call center, manufacturing, supply chain, logistics, healthcare, financial, education
SLIM	MJC2	Strategic logistics modelling and network simulation.	Logistics, manufacturing, supply chain
SLX	Wolverine Software	Large, unique, high-end applications for which there is no off-the-shelf software.	High-end applications requiring end-user customization.
Vanguard Business Analytics Suite	Vanguard Software	Strategic Forecasting, Financial Analysis, Cost Modeling, R&D Pipeline Modeling, Portfolio Analysis, Risk Analysis	All
Vanguard System	Vanguard Software	Strategic Forecasting, Financial Analysis, Cost Modeling, R&D Pipeline Modeling, Portfolio Analysis, Risk Analysis	All
Vector Economics Platform	Vector Economics Inc.	Networking Simulation Applications through use of the DIST standard data type	All Simulation Environments

55 products from 29 vendors!

Simulation tool-based DES Example

Airport Simulation



GPL-based example application

- Consider a **simulation of air traffic arriving and departing at an airport**
- Assume that the airport contains a *single runway* for incoming aircraft
- The simulation might be used to collect *statistics* such as the average number of aircraft waiting to land or the average amount of time each aircraft must wait
 - however, here the focus is on the *event processing mechanism* rather than on the computation of statistics

GPL-based example application

- The **airport state** is characterized by the following state variables:

1. In_The_Air

- indicates the number of aircrafts that are in the process of landing, or are circling, waiting to land

2. On_The_Ground

- indicates the number of aircrafts that have landed and are either at a gate, or traveling to or from a gate

3. Runway_Free

- indicates whether or not the runway is currently being used by a landing aircraft

GPL-based example application

- Three types of events are defined for this simulation:
 1. An **arrival event** denotes the arrival of a new aircraft at the airport
 2. A **landed event** denotes that an aircraft has landed
 3. A **departure event** denotes an aircraft leaving to travel to another airport
- Upon arrival, each aircraft must:
 - a. wait for the runway and land (assume that the aircraft uses the runway for R units of time while landing)
 - b. travel to the gate and unload and load new passengers (assume that this requires G units of time), and
 - c. depart and travel to another airport

Assume that R and G are fixed, known quantities. To simplify the model, queuing at the runway for *departing* aircraft will not be considered here

GPL-based example application

```
/*
```

```
Constants and other values:
```

```
Now = current simulation time (from simulation  
    executive)
```

```
R = time runway in use to land aircraft (constant)
```

```
G = time required at gate (constant)
```

```
State Variables:
```

```
Integer In_The_Air: number of aircrafts landing or  
    waiting to land
```

```
Integer On_The_Ground: number of aircrafts within  
    airport
```

```
Boolean Runway_Free: TRUE if runway is not being used
```

```
Initialize these variables to 0, 0, and TRUE, respectively
```

```
*/
```

Arrival event

New aircraft arrives at airport. If the runway is free, it will begin to land. Otherwise, the aircraft must circle, and wait to land.

- **R** = time runway is used for each landing aircraft
- **G** = time required on the ground before departing
- **Now**: current simulation time
- **InTheAir**: number of aircraft landing or waiting to land
- **OnTheGround**: number of landed aircraft
- **RunwayFree**: Boolean, true if runway available

Arrival Event:

```
InTheAir := InTheAir+1;
```

```
If (RunwayFree)
```

```
    RunwayFree:=FALSE;
```

```
    Schedule Landed event @ Now + R;
```


Landed event

An aircraft has completed its landing.

- **R** = time runway is used for each landing aircraft
- **G** = time required on the ground before departing
- **Now**: current simulation time
- **InTheAir**: number of aircraft landing or waiting to land
- **OnTheGround**: number of landed aircraft
- **RunwayFree**: Boolean, true if runway available

Landed Event:

```
InTheAir:=InTheAir-1;  
OnTheGround:=OnTheGround+1;  
Schedule Departure event @ Now + G;  
If (InTheAir>0)  
    Schedule Landed event @ Now + R;  
Else  
    RunwayFree := TRUE;
```

Departure event

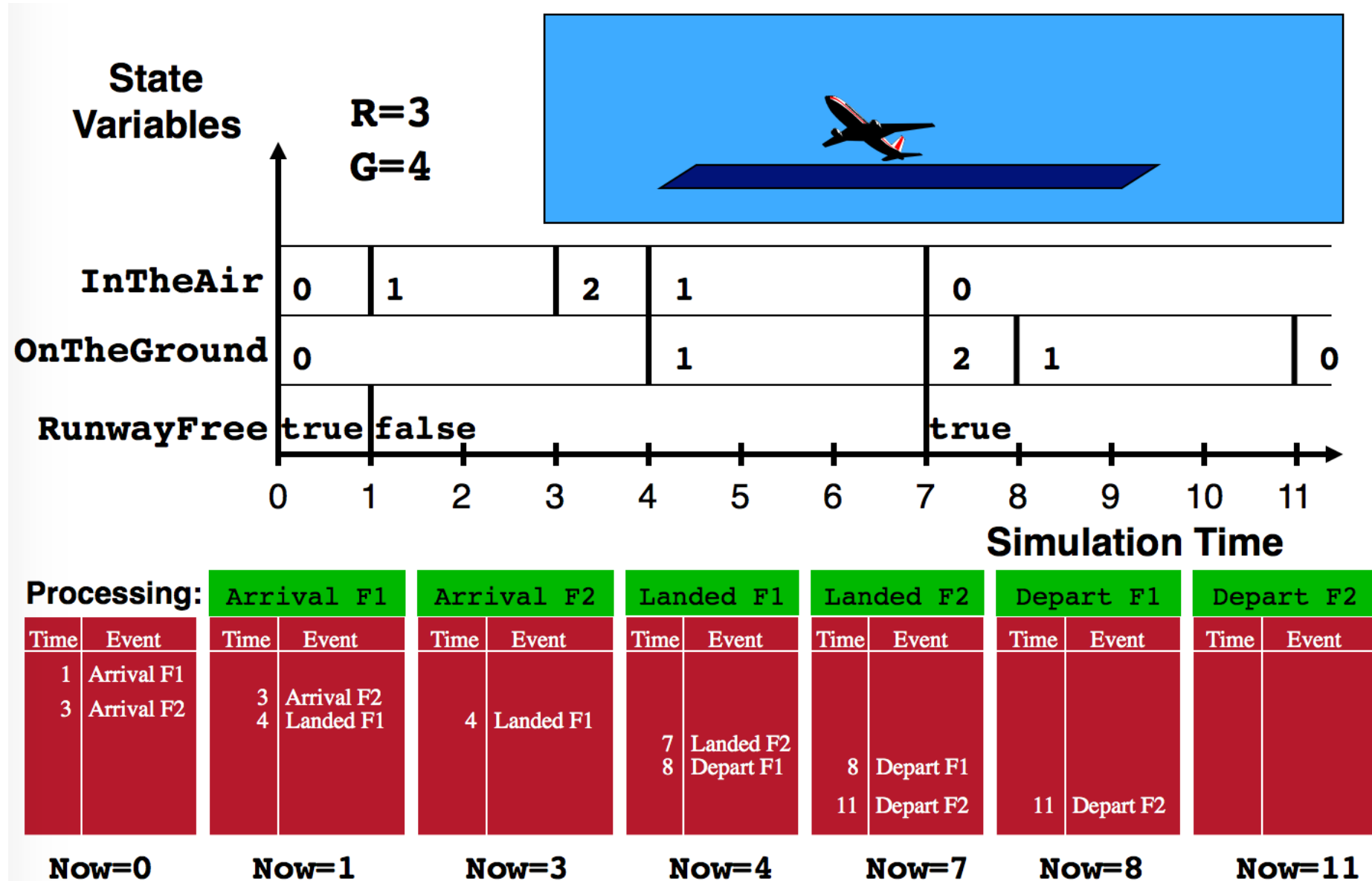
An aircraft now on the ground departs for a new destination.

- **R** = time runway is used for each landing aircraft
- **G** = time required on the ground before departing
- **Now**: current simulation time
- **InTheAir**: number of aircraft landing or waiting to land
- **OnTheGround**: number of landed aircraft
- **RunwayFree**: Boolean, true if runway available

Departure Event:

```
OnTheGround := OnTheGround - 1;
```

Sample program execution



Starting/stopping the simulation

- The **simulation begins** by initializing the state variables, and generating initial events
 - the initial events may be created by defining an "**initialization event**" with time stamp equal to a simulated time prior to the beginning of the actual simulation
 - the simulation application provides a procedure that processes this initialization event by **scheduling all other initial events** required by the simulation.
- There are several techniques for **terminating the execution of the simulation**
 - a "**stop simulation**" event may be used that is defined to be the last event processed by the simulation, even if there are other scheduled events remaining in the event list
 - an "**end simulation time**" may be defined that indicates the simulation is terminated when the simulation clock is about to exceed this time
 - in either case, the simulation will always terminate after an event computation if there are no events in the event list